

西南财经大学本科期末考试试题册（A）

（2025—2026 学年第 1 学期）

以下各项由命题教师填写：

课程名称：数据结构（英） 命题教师：陈中普、王海林
适用对象（年级专业）：2024 级金融数学、2024 级信息管理、2024 级物流管理

使用试题的任课教师姓名：陈中普、王海林

试题说明：

- 1、考试类型：闭卷☒ 开卷☐
- 2、本套试题共 四 道大题，共 6 页，完卷时间 120 分钟。
- 3、考试用品中除纸、笔、尺子外，可另带的用具有：
计算器☐ 字典☐ 等
(请在下划线上填上具体数字或内容，所选☐内打钩)

考试时间（由制卷方填写）：

以下各项由学生填写：

任课教师： 年级专业：
学生姓名： 学 号：

- 考生注意事项：
1. 出示学生证（或身份证）和准考证并放置于桌面左上角，以备查验。
 2. 领取试卷后清点并检查试卷页数，如有重页、页数不足、空白页及印刷模糊等举手向监考教师示意调换试卷。
 3. 答题前先将试题册及答题纸上的任课教师、专业、年级、学号、姓名等信息填写完整。
 4. 所有答案均需填写在答题纸上，填写在试题册上无效。
 5. 考生不得携带任何通讯工具进入考场。
 6. 考试结束后，将试题册、答题纸和草稿纸等材料上交监考教师，不得带离考场。
 7. 严格遵守考场纪律。

By default,

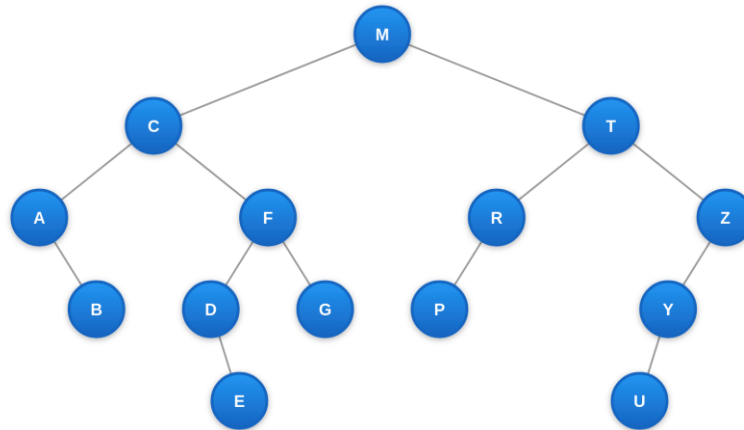
- The height of an empty tree is -1.
- A singly linked list has only its head node as an attribute.
- The node in a singly linked list has attribute data and next, in which data is the element it is holding, and next is the link to the next node.
- A binary tree has only its root node as an attribute.
- The node in a binary tree has attribute key, left, and right, in which key is the data it is holding, and left and right are links to its left and right child, respectively.

Part One: Multi-choices (2 marks each; 30 marks in total)

(Note: Each question has only one correct choice.)

1. Which of the following statements is NOT correct in terms of big O notation?
 - A. $f(N) = N^2 + 3N + 2N\log N$, then $f(N) = N^2$.
 - B. If $f_1(N) = O(N\log N)$ and $f_2(N) = O(N\log N)$, then $f_1(N) = f_2(N)$.
 - C. Given a binary tree with N nodes, we cannot claim that the height of this tree is $O(\log N)$.
 - D. Consider a list in Python, removing the last element can be done in $O(1)$ when using amortized analysis.
2. Given a singly linked list which has only head node, we already implemented `add_first()`, `add_last()`, `remove_first()` and `remove_last()`. How to design an efficient queue?
 - A. `enqueue()` is `add_last()`, while `dequeue()` is `remove_last()`.
 - B. `enqueue()` is `add_first()`, while `dequeue()` is `remove_last()`.
 - C. `enqueue()` is `add_last()`, while `dequeue()` is `remove_first()`.
 - D. `enqueue()` is `add_first()`, while `dequeue()` is `remove_first()`.
3. If an array is used to implement a circular queue, in which *front* is the pointer/index of the top element in the queue, *capacity* is the size of the underlying array, *size* is the number of elements of this queue. Which of the following statements is correct for its `dequeue()` operation?
 - A. *front* will be updated to *front* + 1.
 - B. *front* will be updated to (*front* + 1) % *size*.
 - C. *front* will be updated to (*front* + 1) % *capacity*.
 - D. *front* will be updated to *front* % *capacity* + 1.
4. Given a singly linked list of integers which has only head node, which operation can be done in $O(1)$?
 - A. Get the value of the last node.
 - B. Get the size of the linked list.
 - C. Remove the node with value 10.
 - D. Remove the first node.

5. Given a binary search tree below, how many comparisons are required to check whether it contains letter P?



- A. 3
- B. 4
- C. 5
- D. 6

6. Given the sub-tree rooted at C above, what is sequence of a post-order traversal?

- A. $B \rightarrow A \rightarrow C \rightarrow E \rightarrow D \rightarrow F \rightarrow G$
- B. $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$
- C. $G \rightarrow B \rightarrow A \rightarrow F \rightarrow D \rightarrow E \rightarrow C$
- D. $B \rightarrow A \rightarrow E \rightarrow D \rightarrow G \rightarrow F \rightarrow C$

7. What is the time complexity of the following function in terms of N?

```

def foo(lst):
    N = len(lst) # lst is a Python's list
    total = 0
    for e in range(0, N, 2):
        total += lst[e]
    return total * total
  
```

- A. $O(1)$
- B. $O(N)$
- C. $O(N \log N)$
- D. $O(N^2)$

8. Insert 10, 2, 12, 8, 20, 5, 42, 11 in that order into an initially empty binary search tree using the standard binary search algorithms. In the resulting BST, what is the right child of the node with value 8?

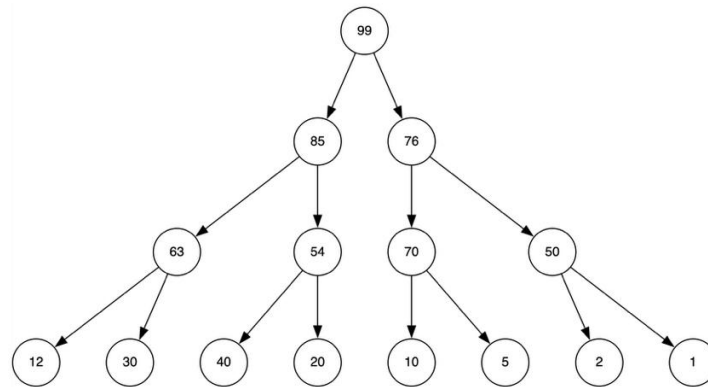
- A. 10
- B. 11
- C. 12
- D. 20

9. Which of the following statements is correct in terms of a red-black tree?

- A. The search operation of a red-black tree is always $O(\log N)$.
- B. For each node x, the height difference of x.left and x.right is no more than 1.
- C. It takes $O(1)$ to remove its root node.

D. The root has no color, and other nodes are either red or black.

10. Given a max-heap below, after deleting the maximum using the standard heap algorithm, where is the node with value 1?



- A. 12's right child.
- B. 30's right child.
- C. 40's right child.
- D. 20's right child.

11. For elements {1, 14, 4, 20, 8, 25, 36, 26, 5}, the hash function is $\text{hash}(x) = x \bmod 10$, and the linear probing is used for open addressing. Suppose elements are stored in an array with 10 slots (from slot 0 to slot 9), what is the element at slot 6?

- A. 5
- B. 25
- C. 26
- D. 36

12. Which of the statements is correct in terms of a hash table?

- A. Hash collision implies an error, so a good hash function should avoid it entirely.
- B. When using the separate chaining, its load factor cannot be greater than 1.
- C. When using the open addressing, its load factor cannot be greater than 1.
- D. Python's dict is based on hash tables, so it is efficient to return all keys greater than a certain value.

13. Given a graph with n nodes and m edges, what is the space cost to store it using an adjacent list?

- A. $O(mn)$
- B. $O(m + n)$
- C. $O(m^2)$
- D. $O(n^2)$

14. Which of the following statement is correct in terms of a graph?

- A. A depth-first search algorithm can be used to compute the shortest path.
- B. A breadth-first search algorithm can be implemented using a queue.
- C. Given an undirected graph with n nodes and m edges, the maximum number of components is $\max(m, n)$.
- D. As for an undirect connected graph with n nodes, if it does not include any parallel edge, then the maximum number of edges is $n(n-1)/2$.

15. In developing a calculator application, it is essential to efficiently evaluate postfix expressions (also known as Reverse Polish Notation, or RPN). For example, the expression "2 3 + 4 *" represents the computation $(2 + 3) * 4 = 20$. In this notation, operands (numbers) appear before their operators, eliminating the need for parentheses and operator precedence rules. Assume the expression includes arbitrary binary operations such as addition (+), subtraction (-), multiplication (*), and division (/). The goal is to achieve an overall time complexity of $O(n)$, where n is the length of the expression, with amortized $O(1)$ time per operation. Which data structure is suitable for this application?

- A. Stack
- B. Queue
- C. Hash table
- D. Tree

Part Two: Blank-filling (2.5 marks each; 25 marks in total)

1. The following pseudo code snippet is to check whether a BST (i.e., *bst*) contains a node with key k . Fill in the blanks in (1)-(4).

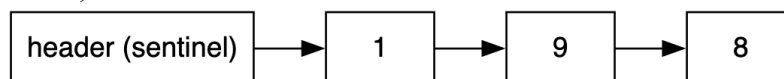
Algorithm isContains(*bst*, k)

```

x ← bst.root
while _____ (1) _____ do
    if _____ (2) _____ then
        x ← x.left
    end
    else
        _____ (3) _____
    end
end
if _____ (4) _____ then
    return False
end
return True

```

2. Suppose we introduce a special *header* node for a singly linked list, which serves as a sentinel at the very first beginning of a linked list, but does not contain any value. Correspondingly, a linked list will only store a header as its attribute. For example, give a linked list $1 \rightarrow 9 \rightarrow 8$, it can be illustrated as:



Please fill the blanks in (5) to finish the pseudo code snippet of removing the first node.

Algorithm remove_first(*lst*)

```

x ← lst.header
if x.next != NULL then
    _____ (5) _____
end

```

3. The following pseudo code snippet is to count the number of nodes in a binary tree (i.e., *tree*) by calling *count_nodes(tree.root)*. Please fill the blanks in (6) and (7).
(Hint: It is a recursive algorithm)

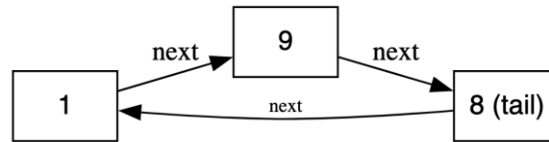
Algorithm count_nodes (node)

```

if node == NULL then
    _____ (6) _____
end
return _____ (7) _____

```

4. In a singly linked list, if the last node (i.e., *tail*) also has a reference to its first node (i.e., *head*), then it is called a *circularly linked list*. It is clear that it is sufficient to store only tail node in a circularly linked list. The following figure depicts $1 \rightarrow 9 \rightarrow 8$.



Please fill the blanks in (8) and (9) to compute the size of a circularly linked list (i.e., *lst*).
(Hint: *lst* has only the tail node, nothing more.)

Algorithm `get_size (lst)`
`tail ← lst.tail`
 if `tail == NULL` then
 `return 0`
 end
`cnt ← 1`
`current ← tail.next`
 while _____ (8) _____ do
 `cnt ← cnt + 1`
 _____ (9) _____
 end
`return cnt`

5. If the load factor is α in a separate-chaining hash table, then a search takes _____(10)_____ time on average, under the assumption that the hash function takes constant time.

Part Three: Short Q&A (5 marks each; 10 marks in total)

1. In data structures, the efficiency of inserting an element at the beginning (*prepending*) varies significantly between Python's built-in list and a singly linked list. This difference stems from their underlying implementations: Python lists are dynamic arrays, while singly linked lists use pointer/link-based nodes.

- Explain why prepending to a Python list incurs $O(N)$ time complexity, where N is the current size of the structure.
- What is the time complexity of prepending operation for a singly linked list?

2.

- Given the max-heap of Q.10 in Part One. Draw the resulting figure after inserting 80.
- Given the sub-tree root at Z of Q.5 in Part One. How to convert it into an AVL tree through rotation? Draw the resulting sub-tree.

Part Four: Analysis and Design (10 + 10 + 15; 35 marks in total)

1. To compute x^N , we need $N-1$ times of multiplication. But we can do better:

- When N is even, $x^N = (x^{N/2}) \times (x^{N/2})$. For example, $x^{10} = x^5 \times x^5$.
- When N is odd, $x^N = (x^{(N-1)/2}) \times (x^{(N-1)/2}) \times x$. For example, $x^{11} = x^5 \times x^5 \times x$.

Based on the observations above, develop a recursive algorithm whose time complexity is $O(\log N)$.

(Hint: You can use pseudo-code, and even plain-English as long as it is clear enough.)

```
def power(x, N) -> int:
    # Your implementation here
```

2. Given functions $f(n)$ and $g(n)$, if there exists real constants c_1 and c_2 , and there are an integer constant $n_0 \geq 1$, such that $c_1 g(n) \leq f(n) \leq c_2 g(n)$ for all $n \geq n_0$, then $f(n) = \Theta(g(n))$. Show that the height of a min-heap with N nodes is $\Theta(\log N)$.

3. There is an actor network graph, in which each actor is a node, and two actors are connected if they collaborated (appeared together) in one movie. As for each actor, it also maintains a list of his/her movies in a singly linked list. In this graph, each action is identified by his/her ID (an integer), and the graph is represented with an adjacent list.

(Hint: You can use pseudo-code, and even plain-English as long as it is clear enough.)

(a) Each movie has attributes including *year*, *title*. A movie list has a *head* node, representing the first movie.

```
class Movie:
    def __init__(self, year, title, next=None):
        self.year = year # int type, year released
        self.title = title # str type
        self.next = next # the link the next node
```

```
class MovieList:
    def __init__(self, head=None):
        self.head = head # the head movie
```

Given a specific movie list *lst*, implement an algorithm to count the number of movies released in the year of 1999. (5 marks)

```
def movie_albums_1999(lst: MovieList) -> int:
    # Your implementation here
```

(b) Given an actor network graph G with an adjacent list *adj*, in which $G.adj[i]$ returns a list/array of actors who were appeared together with the actor i . Additionally, $M[i]$ returns the list of movies in which actor i has appeared.

Given an actor i , implement an algorithm to compute the largest number of movies among i and all other actors who are his/her direct or indirect cooperators. (10 marks)

```
def largest_number_of_movies(G, M, i) -> int:
    # Your implementation here
```